

Solution Requirements

Date	7 July 2026
Team ID	SWTID -2026-4213
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can access the application without login. The system is designed for academic use	Low
2	Authorization levels	Single-user access. No role-based permissions are required in the current version	Low
3	External interfaces	The system integrates with the Wikipedia API for fact verification and Hugging Face Transformer models (DistilBERT and GPT-2) for	High

		event analysis and conversation generatio	
4	Transactions processing	The system processes user inputs (event name, role, and interests), generates personalized conversation starters, verifies facts, and stores history and feedback in JSON files.	High
5	Reporting	The application displays conversation history and user feedback records. Users can review previously generated conversations within the application.	Medium
6	Business rules, etc.	Conversation starters must be generated based on the selected event, user role, and interests. Fact verification should only use reliable Wikipedia information.	High
7	Compliance to laws or regulations	The system stores data locally and does not collect sensitive personal information. It follows basic privacy practices for an academic project	Medium
8	<i>Other</i>	The application provides a simple user-friendly interface with modular architecture for easy maintenance and future enhancements.type your text	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	The system should analyze the event , generate personalized conversation starters, and verify facts quickly without noticeable delay.	AI response generated within 5–10 seconds per request.
2	Scalability	The application should support future integration of additional AI models, databases, and multiple users without major architectural changes.	Modular architecture capable of supporting future multi-user deployment.
3	Security & Data Privacy	User event details, interests, history, and feedback should be stored securely. No sensitive personal information should be shared with unauthorized services.	Local JSON storage with secure API communication and validated user inputs.
4	Reliability & Availability	The application should remain stable while generating conversations, checking facts, and saving history or feedback. It should handle invalid inputs without crashing.	Successful completion of 99% of valid requests with proper error handling.
5	Usability & Accessibility	The interface should be simple and intuitive so that students and professionals can use all features without technical knowledge.	Users should complete the conversation generation process in 3 steps or less with a user-friendly interface.
6	Other	The project should follow a modular structure so that components such as Event Analyzer, Topic Generator , Fact Checker, and Feedback Manager can be updated independently.	Well-organized modules with reusable code and easy future maintenance.